



Tektronix Instruments

How to control Tektronix oscilloscopes, signal generators, and spectrum analyzers with PyVISA. Connection snippets, SCPI commands, and quirks.

Control Tektronix oscilloscopes, AFGs, and spectrum analyzers with PyVISA. One connection pattern, model-specific SCPI.

Supported Models

Category	Series	Examples
Oscilloscopes	TDS/TBS	TDS2000, TBS1000, TBS2000
Oscilloscopes	MSO/DPO	MSO4000, DPO4000, MSO5000, DPO5000
Oscilloscopes	MDO	MDO3000, MDO4000
Signal generators	AFG	AFG3000, AFG31000
Signal generators	AWG	AWG5000, AWG7000
Spectrum analyzers	RSA	RSA5000, RSA6000

Connect to a Tektronix Instrument

```
import pyvisa

rm = pyvisa.ResourceManager()

# USB
```





Supported Models

```
# GPIB
# scope = rm.open_resource("GPIB0::1::INSTR")

scope.timeout = 10000 # 10 s (Tektronix instruments often need longer)
print(scope.query("*IDN?").strip())

scope.write("*RST")
scope.write("*CLS")
```

Log measurements automatically

TofuPilot records test results from your PyVISA scripts, tracks waveform captures and pass/fail rates across production runs. Free to start.

Oscilloscope: Acquire a Waveform

Single-shot acquisition with binary transfer on a TDS/TBS/MSO/DPO scope.

```
import pyvisa
import numpy as np
import time

rm = pyvisa.ResourceManager()
scope = rm.open_resource("TCPIP0::192.168.1.100::inst0::INSTR")
scope.timeout = 10000

# Binary encoding for fast transfer
scope.write("DAT:ENC RIB")
scope.write("DAT:WID 2")      # 16-bit samples
scope.write("DAT:SOU CH1")

# Horizontal and vertical
scope.write("HOR:SCA 1e-6")    # 1 us/div
scope.write("CH1:SCA 0.5")     # 500 mV/div

# Trigger
scope.write("TRIG:MAI:TYP EDGE")
scope.write("TRIG:MAI:EDGE:SOU CH1")
scope.write("TRIG:MAI:LEV 0.1")
scope.write("TRIG:MAI:EDGE:SLO RISE")
```



Supported Models

```
# Wait for trigger
start = time.time()
while time.time() - start < 30:
    if scope.query("ACQ:STATE?").strip() == "0":
```

Built-in Measurements

You can read automatic measurements without transferring the full waveform.

```
# After acquisition completes
scope.write("MEASU:MEAS1:TYP FREQ")
scope.write("MEASU:MEAS1:SOU CH1")
freq = float(scope.query("MEASU:MEAS1:VAL?"))

scope.write("MEASU:MEAS2:TYP AMP")
scope.write("MEASU:MEAS2:SOU CH1")
amp = float(scope.query("MEASU:MEAS2:VAL?"))

scope.write("MEASU:MEAS3:TYP RMS")
scope.write("MEASU:MEAS3:SOU CH1")
rms = float(scope.query("MEASU:MEAS3:VAL?"))

print(f"Frequency: {freq:.1f} Hz, Amplitude: {amp:.3f} V, RMS: {rms:.3f} V")
```

Signal Generator: Output a Waveform

Generate a sine wave on an AFG3000/AFG31000.

```
import pyvisa

rm = pyvisa.ResourceManager()
afg = rm.open_resource("USB0::0x0699::0x0346::C012345::INSTR")
afg.timeout = 5000

afg.write("*RST")

# 1 kHz sine, 2 Vpp, 0.5 V offset
afg.write("SOUR1:FUNC:SHAP SIN")
afg.write("SOUR1:FREQ 1000")
```



Supported Models



```
# Burst mode: 10 cycles per trigger
afg.write("SOUR1:BURS:STAT ON")
afg.write("SOUR1:BURS:MODE TRIG")
afg.write("SOUR1:BURS:NCYC 10")

# Frequency sweep: 100 Hz to 10 kHz over 5 s
afg.write("SOUR1:BURS:STAT OFF")
afg.write("SOUR1:SWE:STAT ON")
afg.write("SOUR1:FREQ:START 100")
afg.write("SOUR1:FREQ:STOP 10000")
afg.write("SOUR1:SWE:TIME 5.0")

afg.close()
rm.close()
```

Spectrum Analyzer: Capture a Trace

Single sweep on an RSA5000/RSA6000.

```
import pyvisa
import numpy as np

rm = pyvisa.ResourceManager()
rsa = rm.open_resource("TCPIP0::192.168.1.100::inst0::INSTR")
rsa.timeout = 30000

rsa.write("*RST")
rsa.write("INIT:CONT OFF") # Single sweep

# 1 GHz center, 100 MHz span, 1 MHz RBW
rsa.write("FREQ:CENT 1e9")
rsa.write("FREQ:SPAN 100e6")
rsa.write("BAND:RES 1e6")
rsa.write("DET:TRAC:FUNC AVER")

# Sweep
rsa.write("INIT:IMM")
rsa.query("*OPC?")

# Read trace
rsa.write("TRAC:DATA? TRACE1")
spectrum = rsa.read_binary_values("f")
```





○ Supported Models



```
peak_idx = np.argmax(spectrum)
print(f"Peak: {freqs[peak_idx]/1e6:.1f} MHz at {spectrum[peak_idx]:.1f} dBm")

rsa.close()
```

Tektronix Quirks

Issue	Detail	Fix
Long timeouts	Tektronix scopes often need 10+ seconds for waveform transfers	Set <code>instrument.timeout = 10000</code> or higher
<code>WFMP?</code> preamble	Comma-separated, field positions vary by firmware version	Print the raw preamble once and verify indices 9, 13, 14 for your model
<code>SYST:ERR?</code> unsupported	Older TDS models don't implement the SCPI error queue	Wrap error checks in <code>try/except</code>
<code>VERB OFF</code>	Terse mode removes headers from query responses, needed for binary parsing	Always send <code>VERB OFF</code> before <code>CURV?</code> transfers
Acquisition state polling	<code>ACQ:STATE?</code> returns <code>"1"</code> while running, <code>"0"</code> when done	Poll with a timeout loop, don't rely on <code>*0PC?</code> alone for single-shot
USB vendor ID	Tektronix USB instruments use vendor ID <code>0x0699</code>	Use this to filter <code>rm.list_resources()</code> results

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